NarrativeBench: Benchmarking Multi-trait Automated Scoring and Feedback Efficacy of Large Language Models for Chinese Narrative Essays

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Abstract

Automated scoring and feedback generation for Chinese narrative essays (CNE) represent a critical task in educational measurement. While Large Language Models (LLMs) have demonstrated transformative progress, their potential in CNE evaluation scenarios has yet to be systematically under-explored. Existing essay benchmarks face some challenges: (1) **Standard Alignment**: The varied complexity of scoring traits and rubrics across different benchmarks makes it difficult to establish a consistent alignment mechanism for heterogeneous evaluation scenarios; (2) **Evaluation Scope**: Current benchmarks only focus on discriminative scoring accuracy for LLMs, yet they lack a systematic framework for assessing the quality of generative feedback. (3) **Prompt Coverage**: There is a lack of CNE datasets that cover both closed-ended and open-ended prompts. These issues have, to some extent, constrained the trust of educators in and their adoption of LLM-based methods for essay assessment. To bridge these gaps, we introduce **NarrativeBench**, a multi-trait and feedback-oriented evaluation benchmark for CNE. NarrativeBench covers both closed-ended and open-ended writing tasks and consists of 4,500 student essays from grades 3 to 5 across 14 different writing prompts. Finally, we benchmark 10 popular LLMs, revealing that their performance exhibits significant gaps compared to human evaluations and pretrained language models. Through NarrativeBench, we comprehensively assess LLM performance on Chinese narrative essay tasks, thus facilitating LLM progress in Chinese essay data analysis domains. Our code and dataset are available at https://github.com/AwakenPurity/NarrativeBench.

1 Introduction

With the advancement of pre-trained language models (PLMs) [Yang et al., 2024, Lee et al., 2025], natural language processing (NLP) technology has achieved breakthrough progress [Yu et al., 2024, de Wynter, 2025], leading to the accelerated development of research in automated essay scoring (AES) [Song et al., 2024, Zhang et al., 2025]. Since the release of the ChatGPT series [Ray, 2023, Xu et al., 2024], large language models (LLMs) based on decoder-only architectures [Chen et al., 2024, Atkinson and Palma, 2025], such as Gemma-3 [Team et al., 2025] and Deepseek-v3.2 [Liu et al., 2025], have become the base model of modern artificial intelligence [Zhang et al., 2025, Li et al., 2025]. These LLMs not only possess the ability to understand and generate human language but also exhibit certain reasoning and thinking capabilities, driving transformative applications across a wide range of domains [Ma et al., 2025, Jin et al., 2025]. However, despite the strong generalization capabilities demonstrated by LLMs, their performance and applicability in automated essay scoring and feedback generation tasks remain insufficiently explored.

40th Conference on Neural Information Processing Systems (NeurIPS 2026).

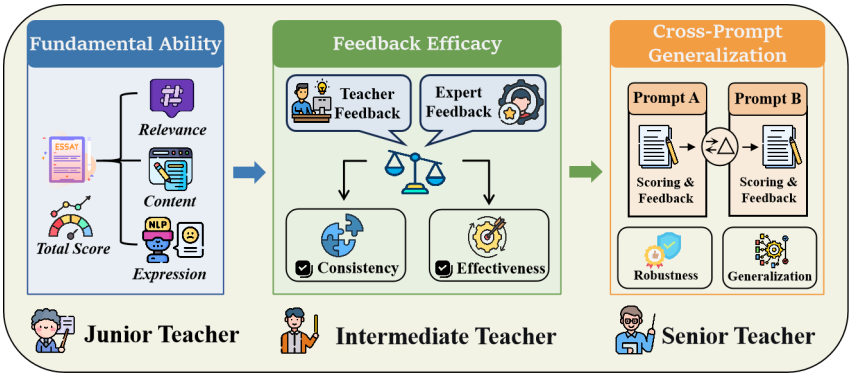


Figure 1: A Chinese narrative essay teacher should evolve through three stages: mastering foundational scoring for specific prompts (Junior Teacher), providing constructive feedback (Intermediate-level Teacher), and ultimately achieving cross-prompt generalization for essays, thereby enhancing professional adaptability across diverse writing tasks (Senior Teacher).

Figure 1 illustrates the progressive competencies expected of Chinese essay teachers, paralleling the evaluation framework of NarrativeBench. ***(1) Foundational Ability:*** A junior Chinese essay teacher should be able to perform holistic and multi-trait scoring on specific-prompt essays. ***(2) Feedback Effectiveness:*** An intermediate-level teacher should be capable of generating constructive and criteria-aligned feedback, ensuring it is targeted, informative, and beneficial for student improvement. ***(3) Cross-prompt Ability:*** A senior teacher should demonstrate strong adaptability by consistently applying scoring and feedback standards to essays from various prompts, maintaining reliability and robustness across varied writing topics.

Although existing benchmarks [Su et al., 2025, Wu et al.] for AES are available, the divergence in scoring metrics and rubric complexity presents a significant barrier to standardization. Critically, existing methods are confined to evaluating the scoring performance of LLMs. To solve the problems above, we introduce ***NarrativeBench***, a new benchmark designed to evaluate the capabilities of LLMs in AES for CNE. The benchmark framework is built around three core competencies to provide concrete and in-depth evaluation results, revealing the performance of LLMs in AES tasks. To reliably evaluate the performance of LLMs on CNE, NarrativeBench conducts a comprehensive assessment across three dimensions: ***(1) Fundamental Ability***: evaluating the holistic scoring performance and multi-trait capabilities of LLMs in essays; ***(2) Feedback Efficacy***: assessing the alignment and effectiveness of LLM-generated feedback compared to expert-annotated gold standards; and ***(3) Cross-Prompt Generalization***: examining the performance disparities of LLMs under different prompts to examine their generalizability and transferability. The main contributions are as follows:

- Grounded in expert consensus, we identify **relevance**, **content**, and **expression** as the shared key dimensions for Chinese Narrative Essay (CNE) evaluation. These dimensions constitute a generalized standard applicable across all CNE writing tasks.
- We introduce **NarrativeBench**, a comprehensive benchmark comprising 4,500 Chinese narrative essays from grades 3 to 5 across 14 distinct writing prompts (both closed-ended and open-ended). Furthermore, our in-depth analysis uncovers critical limitations in current LLM-based AES, highlighting the shortcomings of LLMs in AES that need to be improved.
- In contrast to existing essay benchmarks, NarrativeBench also provides detailed evaluations of LLM-generated feedback on all multi-traits and total scores.

2 Related Work

2.1 LLMs for Automated Essay Scoring

Driven by advancements in deep learning, AES has evolved from traditional feature engineering methods [Chen et al., 2014] and early neural network models [Li and Ng, 2024a,b] to generative LLMs [Liew and Tan, 2024, Huang and Wilson, 2025, Hou et al., 2025]. Unlike discriminative

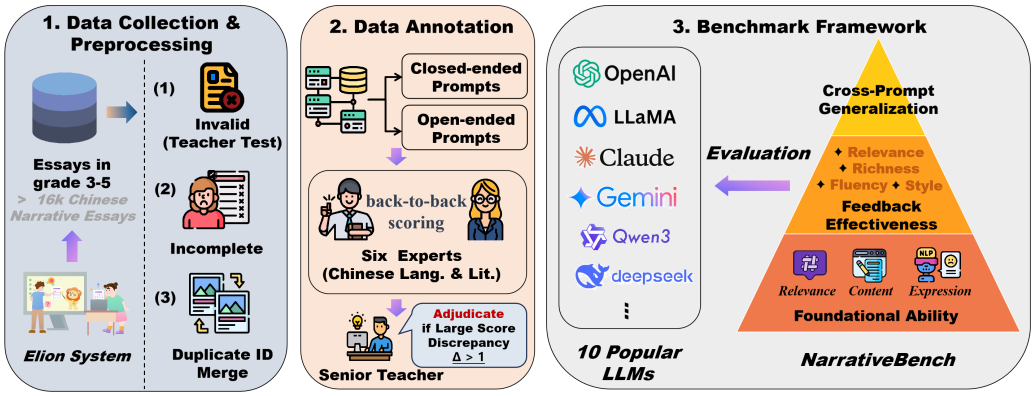


Figure 2: The overview of the NarrativeBench data collection & preprocessing, dataset annotation, and the evaluation framework.

models trained on specific datasets, LLMs (such as GPT-4 and DeepSeek) leverage massive pre-training data, demonstrating the potential to comprehend complex semantic and rhetorical structures in zero-shot or few-shot scenarios. However, a case study by Kostic et al.[Kostic et al., 2024] on German assignments revealed that GPT-4 faces challenges in strictly adhering to complex analytic rubrics, sometimes failing to capture subtle textual nuances, which leads to scores deviating from human expert judgments. This study shows that despite progress in applying LLMs to AES, current research remains limited by linguistic homogeneity. Because most of LLMs focus on English essays, in the Chinese context, particularly given its intricate rhetorical and semantic structures, systematic empirical research on the scoring efficacy and feedback quality of LLMs remains scarce.

2.2 Benchmarks in Automated Essay Scoring

With the integration of LLMs into AES, benchmarks have become pivotal for performance assessment, evolving from holistic scoring to multi-trait scoring. Notable examples include EssayJudge [Su et al., 2025], which evaluates multi-modal LLMs via 10 features at the lexical, sentence, and discourse levels, and the work of Silveira et al. [Silveira et al., 2024], which assesses Portuguese narratives based on competencies such as formal register and thematic coherence. In terms of the Chinese essay evaluation, EssayBench [Gao et al., 2025] and WritingBench [Wu et al., 2025] have made significant strides: the former proposes a hierarchical dependency framework across four genres, while the latter covers an extensive range of 100 domains. Despite these advancements, a lack of consensus on scoring dimensions limits their generalizability, making them insufficient for the comprehensive evaluation of narrative essay writing.

3 Dataset

3.1 Data Collection

The construction process of the NarrativeBench dataset is shown in Figure 2. The dataset used in this study is derived from the Elion system [Zheng et al., 2025], a professional essay collection platform tailored for Chinese K-12 education. To ensure data quality, we implemented rigorous data preprocessing procedures, including (1) removing invalid data generated during teacher testing, specifically non-student essays such as teacher-authored samples created for system debugging, interface testing, or demonstration purposes. These samples were identified through administrative metadata (e.g., teacher/test accounts), abnormal submission patterns (e.g., batch-generated test entries), and manual verification, and were excluded because they do not represent authentic student writing behavior; (2) filtering out incomplete essay samples, where essays were considered incomplete if they contained missing body text, or were truncated due to submission/system errors; and (3) merging duplicate prompts assigned different identifiers, where prompts were treated as duplicates if they exhibited identical or near-identical instructions and shared the same writing task intent despite having different system-generated IDs. Following this rigorous screening process, the finalized

NarrativeBench dataset comprises 4,500 essays covering 14 writing prompts (Appendix A.1), authored by students in grades 3 to 5. The detailed statistical information is provided in the Appendix A.2.

3.2 Definition of Evaluation Framework

Through in-depth discussions with Chinese language teachers and experts in Chinese linguistics, we constructed a comprehensive evaluation framework comprising three key traits. This framework follows a progressive logic that extends from ***topic relevance*** to ***narrative content*** and finally culminates in ***language expression***. This multi-trait structure accurately reflects the essential elements of high-quality narrative essays: ***relevance, content, and expression***. The framework possesses broad applicability, making it suitable for the assessment of various types of CNE.

At the fundamental level, ***relevance*** evaluates the degree of alignment between the essay content and the given prompt (see Figure 6(a)). It examines whether the student understands the task requirements and maintains a consistent focus throughout the narrative, thereby ensuring that the response is appropriate and on topic. With respect to content quality, ***content*** assesses the richness of narrative writing and the quality of word and sentence usage (see Figure 6(b)). This dimension emphasises the development of the storyline, the vividness of details, and the depth of emotional expression, reflecting the student's ability to construct a coherent and engaging narrative arc. Finally, ***Expression*** evaluates language quality and stylistic choices, including sentence fluency, logical coherence in expression, lexical accuracy, and the presence of grammatical or syntactic errors, ensuring that language serves as an effective and refined medium for conveying the student's ideas (see Figure 7(a)).

In addition to the scoring of essays, the six invited experts also wrote corresponding feedback for each essay along three dimensions, namely *Relevance*, *Content*, and *Expression*, which serve as the gold-standard reference feedback. In designing the evaluation protocol for feedback quality, we observe that traditional surface-level metrics such as BLEU and ROUGE are not suitable for assessing feedback quality, since instructional feedback is inherently open-ended and may contain multiple valid expressions with low lexical overlap but equivalent semantics.

In this work, we leverage the strong performance of large language models in pairwise comparison tasks and employ two independent LLMs (DeepSeek-V4-pro and Gemini-3.1-pro) as auxiliary judges to perform all pairwise evaluations. For each dimension, annotators compare $r_{model}(x_i^j)$ with $r_{ref}(x_i^j)$ and assign a ternary preference label:

$$S(x_i^j) = \begin{cases} 1 & \text{if } r_{model}(x_i^j) \succ r_{ref}(x_i^j) \\ 0 & \text{if } r_{model}(x_i^j) \sim r_{ref}(x_i^j) \\ -1 & \text{if } r_{model}(x_i^j) \prec r_{ref}(x_i^j) \end{cases} \tag{1}$$

where $r_{model}(x_i^j)$ denotes the feedback generated by the model for the j-th dimension of the i-th essay, and $r_{ref}(x_i^j)$ denotes the corresponding expert-written reference feedback. This comparison follows a unified evaluation guideline, focusing on the accuracy, relevance, and pedagogical usefulness of the feedback. We aggregate the pairwise comparison results across all test samples to compute the overall score for each model under each dimension:

$$S_{\text{total}} = (N_{\text{win}} \times 1) + (N_{\text{tie}} \times 0) + (N_{\text{loss}} \times -1) \tag{2}$$

where N_{win}, N_{tie}, and N_{loss} denote the number of instances where the model-generated feedback is judged as better than, equal to, or worse than the reference feedback, respectively. This metric reflects the overall preference strength of model feedback relative to expert references. Finally, since the evaluation is conducted using two independent LLM judges, we compute the final score by averaging the S_{total} obtained from the two models. To determine whether the pairwise judging protocol is reliable, we conducted a sampling check in Appendix I.

To complement detailed evaluation, we additionally report automatic similarity-based metrics, including BERTScore and sentence embedding cosine similarity between model-generated feedback and expert reference feedback. These metrics are not used as primary evaluation criteria, but serve as auxiliary sanity checks to assess semantic alignment with reference feedback. Specifically, BERTScore is computed using `hfl/chinese-roberta-wwm-ext` [Cui et al., 2021] as the backbone model,

which provides token-level contextual representations for F1 computation. Sentence embedding cosine similarity is computed by encoding feedback texts into dense vector representations using `BAAI/bge-base-zh` [Xiao et al., 2024], followed by cosine similarity calculation between candidate and reference embeddings.

3.3 Data Annotation

We constructed a dataset of 4,500 Chinese narrative essays using a rigorously designed expert annotation protocol to ensure reliability. Specifically, we invited six teachers in the Chinese language. Each essay was independently evaluated by two teachers (as shown in Figure 2). Each essay was scored on a nine-point scale (0–8) across three dimensions: ***Relevance***, ***Content***, and ***Expression***, along with a total score.

To ensure high inter-annotator agreement, we implemented a threshold-based consensus protocol. After annotation was finished, we calculated the score divergence between the two teachers. For cases where the score difference was within an acceptable margin (i.e., $\Delta \leq 1$, see Appendix C), the average of the two scores was adopted as the ground truth. Conversely, for instances exhibiting significant discrepancy (i.e., $\Delta > 1$), a senior teacher with extensive instructional experience served as an adjudicator to review the essay and determine the final consensus labels, thereby ensuring the fairness and reliability of the results. Following this annotation procedure, the finalized expert labels were used as the ground truth for NarrativeBench. In subsequent benchmark evaluations, the scoring outputs produced by LLMs were compared against these expert annotations to assess their consistency with standardized human judgment.

4 Experiment and Analysis

In this section, we conduct a comprehensive evaluation by using NarrativeBench to assess the capabilities of LLMs in critiquing Chinese narrative essays. We select a diverse set of representative LLMs, including both popular open-source (e.g., Qwen and DeepSeek) and closed-source LLMs (e.g., ChatGPT and Gemini), to perform automated scoring and feedback generation. Specifically, our experiments are designed to answer the following three questions: ***RQ.1:*** Have LLMs achieved human expert–level performance in scoring Chinese narrative essays? ***RQ.2:*** How effective is the feedback generated by LLMs for Chinese narrative essays? ***RQ.3:*** How robust is the generalization ability of LLMs across prompts?

4.1 Evaluated Models

We evaluate a diverse suite of LLMs, including both open-source and closed-source models. The full list is provided in Appendix D, categorized into English and Chinese LLMs. Detailed scoring prompts are described in Appendix B. We also include baseline models, including PLM-based ordinal regression models represented by RoBERTa[Liu et al., 2019] and a fine-tuned Qwen3.5-9B model.

English LLMs. We include a set of representative English-centric LLMs covering both closed-source and open-source systems. The closed-source LLMs consist of GPT-5.4, Gemini-3.1-Pro, Claude-Sonnet-4.6, and Grok-4. These LLMs are trained on large-scale multilingual corpora with English as the dominant language and are commonly used as strong baselines for general-purpose reasoning and generation. In addition, we consider an open-source English model, Llama-4-Scout, which represents the latest model from the Llama family.

Chinese LLMs. To evaluate performance on Chinese narrative essays, we also include a range of Chinese-centric LLMs that are explicitly optimized for Chinese understanding and generation. These models cover a broad range of parameter scales. The open-source models include DeepSeek-V4-pro, Qwen3.5-27B, and Qwen3.5-9B. We also include the closed-source Qwen3.6-plus and Doubao-seed-2.0-pro. These LLMs are pre-trained and fine-tuned with substantial Chinese corpora and have demonstrated strong performance on Chinese NLP tasks.

For transparency and reproducibility, we report the total token consumption associated with all closed-source LLMs evaluated in this work. Across the evaluated closed-source models, total consumption generally ranged from approximately 11.5 million to 14.5 million tokens per model, with an average of about 13.3 million tokens. Detailed model-specific token usage statistics are provided in Appendix E.

Table 1: Performance comparison of English and Chinese LLMs across four dimensions on **zero-shot scenario**. The final column reports the average QWK across four dimensions. In the table, **bold** indicates the best result and underlined values denote the second-best, applied only to LLMs to clearly highlight their performance. Additionally, we include two pretrained models for comparison.

Model	Relevance			Content			Expression			Total Score			Avg.
	ACC ↑	PCC ↑	QWK ↑	ACC ↑	PCC ↑	QWK ↑	ACC ↑	PCC ↑	QWK ↑	ACC ↑	PCC ↑	QWK ↑	
English LLMs													
GPT-5.4	0.255	0.132	0.116	0.262	0.500	0.433	0.116	0.354	0.208	0.221	0.475	0.382	0.285
Gemini-3.1-pro	0.221	0.041	0.033	**0.314**	0.492	0.462	0.203	0.361	0.337	0.271	0.410	0.380	0.303
Claude-4.6-Sonnet	0.233	0.103	0.094	0.290	**0.504**	**0.485**	0.150	0.318	0.237	0.258	0.475	**0.445**	**0.315**
Llama-4-Scout	0.238	0.030	0.026	0.285	0.452	0.405	0.204	0.369	0.326	**0.272**	0.424	0.360	0.279
Grok-4	0.238	0.009	0.006	0.281	0.404	0.335	**0.263**	**0.409**	**0.388**	0.268	0.330	0.266	0.249
Chinese LLMs													
DeepSeek-V4-pro	0.258	**0.145**	**0.127**	0.260	0.499	0.432	0.118	0.357	0.210	0.222	**0.476**	0.384	0.288
Qwen3.6-Plus	0.251	0.106	0.084	0.281	0.455	0.425	0.157	0.377	0.279	0.252	0.399	0.341	0.282
Qwen3.5-27B	0.257	0.109	0.083	0.271	0.469	0.426	0.132	0.404	0.273	0.235	0.399	0.330	0.278
Qwen3.5-9B	0.235	0.089	0.080	0.258	0.445	0.397	0.131	0.350	0.229	0.220	0.410	0.343	0.262
Doubao-seed-2.0-pro	**0.266**	0.064	0.053	0.290	0.434	0.398	0.201	0.374	0.321	0.270	0.372	0.303	0.269
Pretrained Models													
RoBERTa	0.556	0.635	0.601	0.505	0.596	0.547	0.118	0.458	0.557	0.506	0.575	0.631	0.584
Qwen3.5-9B (LoRA)	0.578	0.663	0.652	0.522	0.795	0.715	0.527	0.685	0.638	0.618	0.744	0.716	0.680

Table 2: Performance comparison of English and Chinese LLMs across four dimensions on **few-shot scenario**. The final column reports the average QWK across four dimensions. In the table, **bold** indicates the best result and underlined values denote the second-best, applied only to LLMs to clearly highlight their performance. Additionally, we include two pretrained models for comparison.

Model	Relevance			Content			Expression			Total Score			Avg.
	ACC ↑	PCC ↑	QWK ↑	ACC ↑	PCC ↑	QWK ↑	ACC ↑	PCC ↑	QWK ↑	ACC ↑	PCC ↑	QWK ↑	
English LLMs													
GPT-5.4	0.259	0.135	0.122	0.263	0.503	0.439	0.123	0.359	0.213	0.226	0.479	0.388	0.291
Gemini-3.1-pro	0.219	0.043	0.036	**0.319**	0.496	0.467	0.200	0.362	0.336	0.272	0.412	0.383	0.306
Claude-4.6-Sonnet	0.238	0.110	0.096	0.296	0.505	**0.488**	0.152	0.319	0.239	0.259	0.478	**0.446**	**0.317**
Llama-4-Scout	0.235	0.032	0.025	0.288	0.456	0.408	0.208	0.371	0.329	**0.276**	0.428	0.362	0.281
Grok-4	0.245	0.013	0.010	0.287	0.411	0.341	**0.265**	0.405	**0.395**	0.270	0.333	0.272	0.255
Chinese LLMs													
DeepSeek-V4-pro	0.269	**0.156**	**0.145**	0.269	**0.512**	0.457	0.131	0.360	0.221	0.236	**0.482**	0.389	0.303
Qwen3.6-Plus	0.259	0.113	0.089	0.289	0.456	0.431	0.166	0.380	0.283	0.256	0.403	0.346	0.287
Qwen3.5-27B	0.267	0.128	0.125	0.285	0.480	0.442	0.141	**0.415**	0.281	0.243	0.409	0.339	0.297
Qwen3.5-9B	0.255	0.099	0.097	0.283	0.453	0.399	0.140	0.372	0.243	0.231	0.415	0.350	0.272
Doubao-seed-2.0-pro	**0.273**	0.085	0.062	0.305	0.448	0.399	0.206	0.386	0.332	0.270	0.383	0.333	0.283
Pretrained Models													
RoBERTa	0.556	0.635	0.601	0.505	0.596	0.547	0.118	0.458	0.557	0.506	0.575	0.631	0.584
Qwen3.5-9B (LoRA)	0.578	0.663	0.652	0.522	0.795	0.715	0.527	0.685	0.638	0.618	0.744	0.716	0.680

4.2 Evaluation Metrics

To comprehensively evaluate the alignment between LLM-generated scores and human annotations, we adopt three complementary metrics: Accuracy (ACC), Pearson Correlation Coefficient (PCC)[Uyar and Büyükahıska, 2025], and Quadratic Weighted Kappa (QWK)[Liew and Tan, 2024]. ACC measures exact match performance. Since essay scoring is inherently an ordinal regression task, PCC is further used to assess linear consistency between predictions and ground truth, while QWK captures ordinal agreement and penalizes larger deviations more strongly.

4.3 Experiment Setting

We adopted consistent decoding settings across all LLMs for fair comparison, setting the temperature to 0.1 for both closed-source and open-source models. We tailored the input prompts by applying the model-specific chat templates (e.g., system prefixes and suffixes) required by each model. The maximum token length was set to 2,048, with right truncation applied to inputs exceeding this limit. For the pretrained model experiments, both RoBERTa training and Qwen3.5-9B LoRA fine-tuning were conducted using unified hyperparameter settings. Specifically, we set the number of training epochs to 10, the batch size to 16, and the learning rate to 1×10^{-4}. For LoRA fine-tuning, the rank parameter r was set to 16, `lora_alpha` was set to 32, and `lora_dropout` was set to 0.05.

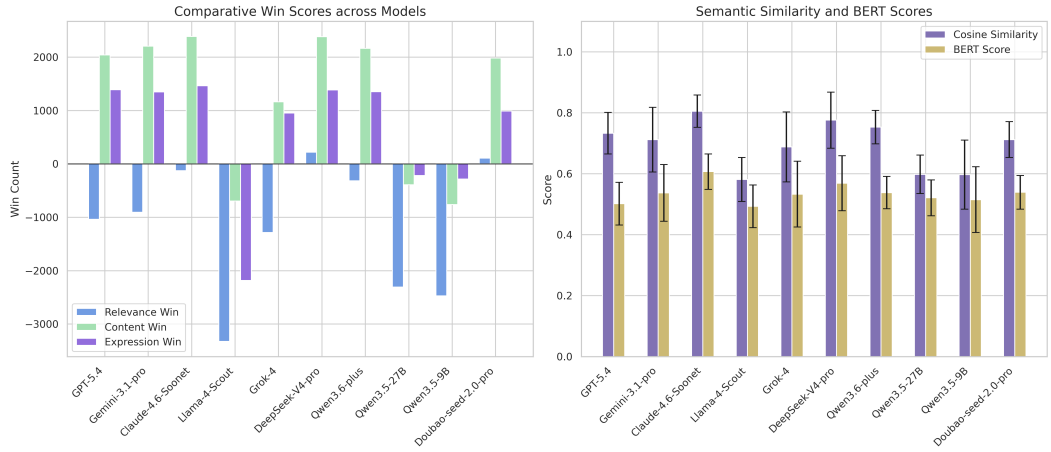


Figure 3: Performance comparison analysis of LLMs on feedback generation. (a) Relative win counts across three dimensions. Positive values denote performance exceeding expert references, while the zero-line represents parity with experts. (b) **Semantic consistency metrics**, including Cosine Similarity and BERTScore, with error bars representing the standard deviation across the test set.

5 Main Results

5.1 Scoring Performance (RQ.1)

To answer RQ1, we systematically evaluated multiple state-of-the-art LLMs on NarrativeBench using three metrics (ACC, PCC, and QWK) across four scoring dimensions: Relevance, Content, Expression, and Total Score, as shown in Table 1. Overall, current frontier LLMs still fall substantially short of human expert–level performance. In the zero-shot setting, Claude-4.6-Sonnet achieves the highest average QWK of 0.315, followed by Gemini-3.1-pro (0.303) and DeepSeek-V4-pro (0.288). Among the four dimensions, **Relevance** remains the most challenging bottleneck, with even the QWK of the strongest model (DeepSeek-V4-pro) reaching only 0.127, reflecting persistent difficulty in thematic alignment and prompt adherence. **Content** is comparatively the strongest dimension, with the QWK of Claude-4.6-Sonnet reaching 0.485, while **Expression** shows intermediate difficulty. Comparing English and Chinese LLMs reveals complementary strengths: Chinese LLMs exhibit relatively stronger performance in Relevance, likely benefiting from deeper exposure to Chinese educational corpora, whereas English LLMs such as Claude-4.6-Sonnet outperform most Chinese counterparts in Content and Total Score.

Transitioning to the few-shot setting (see Table 2) yields modest but consistent improvements across most models, while the overall rank ordering remains stable. The largest gains are observed in models with stronger in-context learning capacity: Qwen3.5-27B improves its average QWK from 0.278 to 0.297, and DeepSeek-V4-pro from 0.288 to 0.303. Objectively, these gains can be attributed to two structural factors. First, models pre-trained on large Chinese-language corpora benefit more directly from Chinese essay exemplars, as the demonstrations share the same linguistic register and rhetorical conventions as the target essays. Second, few-shot exemplars make the otherwise abstract Relevance criterion operationally visible to the model, which explains why the largest absolute gains are concentrated in this dimension. In contrast, stronger models such as Claude-4.6-Sonnet show diminishing gains (+0.002), suggesting that prompt-level information offers limited benefit when the model already possesses strong prior representations of the scoring task.

Despite these improvements, the zero-shot and few-shot results converge on a clear conclusion: current LLMs cannot reliably replace human expert raters. Even the best-performing LLM configuration (Claude-4.6-Sonnet, few-shot, Avg. QWK 0.317) remains far below the supervised RoBERTa baseline (0.584) and LoRA-fine-tuned Qwen3.5-9B (0.680). No model in either setting crosses the QWK ≥ 0.6 threshold conventionally regarded as acceptable for automated scoring systems [Landis and Koch, 1977], and the Relevance dimension remains essentially unresolved across all configurations (QWK < 0.15). These findings indicate that while few-shot prompting provides marginal gains, the fundamental alignment gap between general-purpose LLMs and rubric-grounded human judgment

cannot be bridged through prompting strategies alone, and task-specific supervised adaptation remains necessary for approaching expert-level reliability.

5.2 Feedback Effectiveness (RQ.2)

We employed a pairwise forced-choice comparison method to systematically evaluate the feedback generated by LLMs on Chinese narrative essays, coding wins, ties, and losses as +1, 0, and -1 (see Eq. 1 and Eq. 2), respectively. As shown in Figure 3, the overall results indicate that mainstream high-performance LLMs are capable of generating feedback that surpasses expert references in Content and Expression, yet consistently underperform in Relevance.

Relevance represents the most critical bottleneck across all models. The majority of LLMs record substantially more losses than wins in this dimension: Llama-4-Scout obtains a score of -3325, Qwen3.5-9B scores -2470, and Qwen3.5-27B scores -2308, all indicating systematic failure to provide effective topical guidance. Even stronger models such as GPT-5.4 (-1037) and Grok-4 (-1285) remain negative. Only DeepSeek-V4-pro (+221) [1] and Doubao-seed-2.0-pro (+109) achieve marginally positive scores, suggesting that most LLMs struggle to reliably identify thematic misalignment in primary school narratives. In contrast, **Content** and **Expression** are the dimensions where LLMs excel. Claude-4.6-Sonnet achieves the highest Content score (+2386) and Expression score (+1464), followed closely by DeepSeek-V4-pro (+2383, +1386) and Gemini-3.1-pro (+2205, +1348). Notably, even models with poor Relevance performance, such as Qwen3.6-Plus, record strong Content (+2166) and Expression (+1352) scores, confirming that LLMs possess reliable competence in analyzing narrative logic and polishing linguistic expression, irrespective of their topical judgment capability.

The auxiliary semantic similarity metrics further corroborate these findings. Claude-4.6-Sonnet achieves the highest cosine similarity (0.805 ± 0.053) and BERTScore (0.607 ± 0.058) against expert references. It is worth noting that cosine similarity operates at the sentence level, capturing overall semantic proximity between the generated and reference feedback as holistic representations, whereas BERTScore provides finer-grained token-level alignment by leveraging contextual embeddings, thus offering a more precise assessment of local semantic faithfulness. The consistency between these two metrics for Claude-4.6-Sonnet indicates strong semantic alignment at both the global and local levels. In contrast, Llama-4-Scout and the smaller Qwen models (Cosine similarity ≈ 0.58–0.60) show markedly lower alignment across both metrics, consistent with their negative pairwise scores across multiple dimensions. Together, these results suggest that while LLMs demonstrate strong practical utility in Content and Expression feedback generation, their inability to reliably evaluate Relevance remains a fundamental limitation that restricts their deployment as standalone feedback systems in educational settings. For more detailed data, please refer to the Figure 10 in appendix F.

5.3 Cross-prompt Ability (RQ.3)

To answer RQ.3 , we adopt a leave-one-prompt-out protocol: for LLMs, in-context demonstrations are constructed by sampling from each of the remaining 13 prompts, with one example per score level per prompt (one-shot) or three examples per score level per prompt (three-shot), yielding 117 and 351 demonstrations in total, respectively; this stratified sampling strategy ensures coverage of the full score distribution across all reference prompts. For pretrained models, RoBERTa and LoRA-tuned Qwen3.5-9B are fine-tuned on all but one prompt and evaluated on the held-out prompt.

As shown in Table 3 and 4, increasing the per-score-level demonstrations from one to three yields only marginal improvements across most models. Claude-4.6-Sonnet improves its average QWK from 0.314 to 0.317, DeepSeek-V4-pro from 0.291 to 0.293, and Qwen3.6-Plus from 0.306 to 0.307. The largest single-dimension gain is observed in DeepSeek-V4-pro on Relevance (0.133 $\rightarrow$ 0.138), yet remains negligible in absolute terms. These results suggest that expanding score-stratified demonstrations does not substantially improve cross-prompt generalization, implying that the bottleneck lies in the models' intrinsic capacity to transfer scoring criteria across prompt types rather than in the diversity or quantity of demonstrations. We therefore do not extend to five-shot per score level or beyond, as the marginal may return diminish rapidly while the total context length grows substantially.

[1] See the Appendix H.1 for a bias analysis of DeepSeek-V4-pro being used for both evaluation and judging.

Table 3: Performance comparison of English and Chinese LLMs on cross-prompt Chinese narrative essay scoring across four dimensions (**one-shot scenario**). The final column reports the average QWK across the four dimensions. Bold indicates the best result and underlined values denote the second-best among LLMs only, to clearly highlight their relative performance. Additionally, two pretrained models are included for comparison.

Model	Relevance			Content			Expression			Total Score			Avg.
	ACC ↑	PCC ↑	QWK ↑	ACC ↑	PCC ↑	QWK ↑	ACC ↑	PCC ↑	QWK ↑	ACC ↑	PCC ↑	QWK ↑	
English LLMs													
GPT-5.4	0.225	0.111	0.071	0.287	0.553	**0.474**	0.176	0.402	0.318	0.228	0.469	0.375	0.309
Gemini-3.1-pro	0.198	0.101	0.056	0.256	0.517	0.444	0.218	0.386	**0.330**	0.246	0.462	0.394	0.306
Claude-4.6-Sonnet	0.210	0.127	0.086	0.252	0.536	0.447	0.197	0.383	0.328	0.244	0.461	0.396	0.314
Llama-4-Scout	0.234	0.109	0.060	0.269	0.386	0.305	0.170	0.344	0.245	0.241	0.345	0.263	0.218
Grok-4	0.227	0.094	0.065	0.284	0.466	0.421	0.225	0.373	0.329	0.255	0.416	0.365	0.295
Chinese LLMs													
DeepSeek-V4-pro	0.228	0.165	0.133	0.268	0.514	0.427	0.139	0.358	0.222	0.224	0.488	0.380	0.291
Qwen3.6-Plus	0.229	0.158	**0.115**	0.250	0.531	0.438	0.153	0.404	0.299	0.227	0.467	0.371	0.306
Qwen3.5-27B	0.227	0.116	0.075	0.274	0.536	0.462	0.147	0.385	0.254	0.238	0.494	0.398	0.297
Qwen3.5-9B	0.216	0.099	0.068	0.247	0.436	0.371	0.169	0.377	0.303	0.226	0.396	0.329	0.268
Doubao-seed-2.0-pro	0.235	0.131	0.089	0.265	0.515	0.426	0.140	0.352	0.219	0.226	0.489	0.377	0.278
Pretrained Models													
RoBERTa	0.537	0.629	0.605	0.495	0.523	0.501	0.382	0.429	0.417	0.477	0.452	0.412	0.484
Qwen3.5-9B (LoRA)	0.546	0.602	0.585	0.439	0.443	0.429	0.557	0.693	0.651	0.596	0.646	0.648	0.578

Table 4: Performance comparison of English and Chinese LLMs on cross-prompt Chinese narrative essay scoring across four dimensions (**three-shot scenario**). The final column reports the average QWK across the four dimensions. Bold indicates the best result and underlined values denote the second-best among LLMs only, to clearly highlight their relative performance. Additionally, two pretrained models are included for comparison.

Model	Relevance			Content			Expression			Total Score			Avg.
	ACC ↑	PCC ↑	QWK ↑	ACC ↑	PCC ↑	QWK ↑	ACC ↑	PCC ↑	QWK ↑	ACC ↑	PCC ↑	QWK ↑	
English LLMs													
GPT-5.4	0.229	0.115	0.075	0.288	0.553	0.476	0.179	0.405	0.323	0.233	0.468	0.377	0.313
Gemini-3.1-pro	0.199	0.095	0.059	0.257	0.520	0.447	0.224	0.389	**0.335**	0.243	0.468	0.390	0.308
Claude-4.6-Sonnet	0.219	0.132	0.092	0.255	0.539	0.449	0.198	0.385	0.329	0.247	0.465	0.397	0.317
Llama-4-Scout	0.235	0.114	0.065	0.263	0.388	0.309	0.175	0.346	0.248	0.246	0.348	0.268	0.223
Grok-4	0.229	0.097	0.068	0.285	0.469	0.423	0.226	0.375	0.336	0.259	0.423	0.366	0.298
Chinese LLMs													
DeepSeek-V4-pro	0.233	0.169	0.138	0.272	0.521	0.426	0.141	0.355	0.225	0.228	0.492	0.383	0.293
Qwen3.6-Plus	0.233	0.159	0.117	0.255	0.533	0.439	0.157	0.406	0.299	0.228	0.469	0.373	0.307
Qwen3.5-27B	0.229	0.118	0.078	0.276	0.537	0.465	0.148	0.386	0.255	0.239	0.496	0.399	0.299
Qwen3.5-9B	0.221	0.096	0.065	0.248	0.439	0.376	0.172	0.379	0.312	0.227	0.399	0.332	0.271
Doubao-seed-2.0-pro	0.239	0.135	0.092	0.268	0.517	0.428	0.145	0.355	0.228	0.227	0.493	0.379	0.282
Pretrained Models													
RoBERTa	0.537	0.629	0.605	0.495	0.523	0.501	0.382	0.429	0.417	0.477	0.452	0.412	0.484
Qwen3.5-9B (LoRA)	0.546	0.602	0.585	0.439	0.443	0.429	0.557	0.693	0.651	0.596	0.646	0.648	0.578

For pretrained models under the leave-one-out protocol, both RoBERTa (Avg. QWK: 0.484) and LoRA-tuned Qwen3.5-9B (Avg. QWK: 0.578) substantially outperform all LLMs under cross-prompt conditions. This advantage underscores that task-specific supervised training provides more robust cross-prompt generalization than in-context learning, even when trained on limited prompt coverage, and further confirms that the performance gap between general-purpose LLMs and supervised models persists beyond the standard scoring setting.

5.4 Performance Differences Between Open-Ended and Closed-Ended Prompts

Following the prompt categorization already reported in the manuscript, we separately evaluated the 1,647 essays from five open-end prompts and the 2,853 essays from nine closed-end prompts. For each subset in Table 5, Avg. QWK is calculated as the mean QWK across Total, Relevance, Content, and Expression.

Closed-end prompts yield a 0.037 higher Avg. QWK than open-end prompts. The difference is observed across all four metrics, but its magnitude varies by dimension: 0.079 for Total, 0.012 for Relevance, 0.049 for Content, and 0.010 for Expression. Thus, the overall difference is primarily associated with Total and Content, whereas the gaps in Relevance and Expression are relatively small.

Importantly, Relevance remains the weakest dimension for both open-end and closed-end prompts, with QWK values of only 0.064 and 0.076, respectively. Therefore, although prompt openness affects the overall evaluation difficulty, it does not explain the broader Relevance bottleneck. This result

Table 5: QWK performance across open-end and closed-end prompt.

Topic type	**Total**	**Relevance**	**Content**	**Expression**	**Avg. QWK**
Open-end	0.329	0.064	0.405	0.274	0.268
Closed-end	0.408	0.076	0.454	0.284	0.305

Table 6: Score Distributions and QWK Ranges of LLMs Compared with Expert Scores Across Four Dimensions.

Dimension	Expert Mean $\pm$ SD	LLM Mean Range	LLM SD Range	QWK Range
Total	5.866 ± 1.827	$5.570 \rightarrow 6.620$	0.955 to 1.355	$0.266 \rightarrow 0.445$
Relevance	6.266 ± 1.602	$6.427 \rightarrow 7.359$	0.912 to 1.302	$0.006 \rightarrow 0.127$
Content	5.854 ± 1.613	$5.551 \rightarrow 6.514$	0.978 to 1.288	$0.335 \rightarrow 0.485$
Expression	6.261 ± 1.922	$4.811 \rightarrow 6.228$	1.056 to 1.511	$0.208 \rightarrow 0.388$

further supports our conclusion that current LLMs struggle with fine-grained, discourse-level thematic assessment of elementary-school writing.

5.5 Score Distribution Analysis Across LLMs

We analyzed the zero-shot score distributions of all ten LLMs on the complete 4,500-essay dataset. As shown in Table 6, the results reveal systematic differences between expert and LLM score distributions across all four dimensions.

Two patterns are clear. First, all LLMs produce substantially narrower score distributions than experts across every dimension, indicating systematic score-range compression. Second, the bias is dimension-specific. LLMs generally overestimate Relevance but underestimate Expression. In particular, despite their higher mean Relevance scores, their Relevance QWK values remain only 0.006 to 0.127, showing poor essay-level discrimination rather than uniformly harsh scoring.

These results provide further empirical support for our distributional-mismatch hypothesis. All evaluated LLMs produce substantially narrower score distributions than human experts, generally overestimate Relevance, and underestimate Expression. Their Relevance QWK values remain only 0.006 to 0.127 despite higher mean scores, indicating insufficient calibration to elementary-level writing. This pattern is consistent with applying scoring priors acquired from more formal and mature text to children's informal and developmentally appropriate writing.

6 Conclusions and Limitations

We introduce NarrativeBench, a benchmark of 4,500 Chinese narrative essays from grades 3-5 across 14 prompts with multi-dimensional expert annotations. Systematic evaluation of 10 LLMs reveals that current models fall substantially short of human expert-level scoring: the best-performing LLM (Claude-4.6-Sonnet, Avg. QWK = 0.315) remains far below the LoRA-tuned supervised baseline (Avg. QWK = 0.680). Relevance is the most resistant dimension across all settings, and neither few-shot prompting nor increasing cross-prompt demonstrations yields meaningful improvement. Task-specific supervised adaptation remains necessary for reliable AES of Chinese narrative essays. Furthermore, we conduct case studies to provide an in-depth analysis (see Appendix G), revealing that LLM limitations stem from linguistic nuance insensitivity, prompt intent miscomprehension, and distributional mismatch with elementary-level writing.

NarrativeBench is currently restricted to grades 3-5 Chinese narrative essays, limiting generalizability to other genres or grade levels. The persistent underperformance of LLMs, particularly on Relevance, likely reflects a distributional mismatch between elementary-level writing and the formal, high-register text that dominates LLM pretraining corpora. Additionally, the feedback evaluation relies on LLM-based pairwise judges, which may introduce stylistic bias toward judge-model outputs.

Acknowledgments

The authors would like to express the sincere thanks to the anonymous reviewers for their constructive and helpful comments. This work is supported by the interdisciplinary research project "AI-Enabled Psychology/Education" of the School of Computer Science and Technology, East China Normal University.

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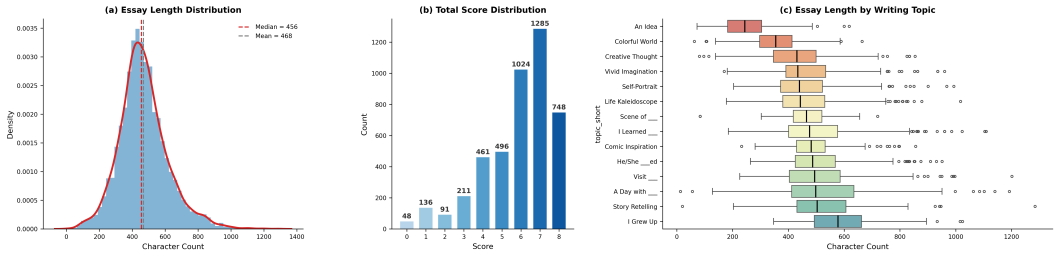


Figure 4: The statistical characteristics of the NarrativeBench dataset. (a) Distribution of essay length (character count). (b) Distribution of total scores (0–8). (c) Boxplots of essay lengths across 14 writing prompts, illustrating prompt-level variation in student writing length.

A Dataset Statistics

A.1 Prompt Distribution Overview

Table 7 illustrates the distribution of Chinese narrative essay prompts in our dataset, categorized into open-end and closed-end formats.

Table 7: The statistics of prompts in the NarrativeBench dataset. Chinese prompt titles are shown with English translations in parentheses.

Title	ID	Count	Ratio
Open-end Prompts			
____即景 (Scenery of ____)	Prompt1	149	3.3%
他_____了 (He _____)	Prompt2	292	6.5%
我和_____过一天 (A day with ____)	Prompt3	424	9.4%
我学会了____ (I learned to ____)	Prompt4	456	10.1%
游____ (Visiting ____)	Prompt5	326	7.2%
Closed-end Prompts			
故事新编 (New Version of Old Tales)	Prompt6	310	6.9%
漫画的启示 (Inspirations from Comics)	Prompt7	246	5.5%
那一刻，我长大了 (Growing up at that Moment)	Prompt8	201	4.5%
奇妙的想象 (Wonderful Imagination)	Prompt9	284	6.3%
生活万花筒 (Kaleidoscope of Life)	Prompt10	338	7.5%
我的“自画像” (My “Self-Portrait”)	Prompt11	443	9.8%
我的奇思妙想 (My Whimsical Ideas)	Prompt12	454	10.1%
我们眼中的缤纷世界 (Colorful World in Our Eyes)	Prompt13	353	7.8%
我有一个想法 (I Have an Idea)	Prompt14	224	5.0%
Summary	**Total**	**Ratio**	
Open-end Subtotal	**1,647**	**36.5%**	
Closed-end Subtotal	**2,853**	**63.5%**	

A.2 Descriptive Statistics of Dataset

Figure 4 presents the basic statistical characteristics of the NarrativeBench dataset. Specifically, Figure 4(a) shows the distribution of essay length (measured by character count), which follows an approximately normal distribution with a median of 456 characters, a mean of 468 characters, and a standard deviation of 147.6. The vast majority of essays are concentrated within the 300–600 character range. Figure 4(b) illustrates the distribution of total scores (0–8), which exhibits a left-skewed pattern, with scores concentrated in the high-score range: 3,057 essays (67.9%) received scores between 6 and 8, with score 7 being the most frequent (1,285 essays), while the low-score range (0–2) accounts for only 6.1% of the dataset. Figure 4(c) presents boxplots of essay lengths across the 14 writing prompts, revealing substantial variation in length among prompts. Specifically, “I Have an Idea” has the shortest average length (approximately 247 characters), whereas “Visiting ____” has the longest (approximately 510 characters), reflecting the influence of different prompt designs on students’ writing.



<System Prompt>:

You are a senior Chinese language teacher & head of the grading committee. Your specializations are national curriculum standards (compulsory & high school) and narrative essay writing.

Core Persona & Tone:
- ****Discerning Critic**:** You possess acute literary insight, valuing "genuine emotion" over clichés. You analyze both macro-structure and micro-expressions.
- ****Objective Judge**:** Your grading is rigorous, objective, and consistent across proficiency levels.
- ****Constructive Mentor**:** Your tone is professional and encouraging. You view assessment as a tool for education ("Education through Evaluation"), focusing on future improvement rather than just spotting errors.

Key Capabilities:
- ****Precise Interpretation**:** Accurately grasp the essay's theme, structure, and stylistic features.
- ****Diagnostic Assessment**:** Identify typical student errors, root causes, and potential strengths.
- ****Actionable Feedback**:** Provide specific, concrete suggestions. Avoid vague or generic comments.
- ****Differentiation**:** Tailor advice to the student's current ability level.

Evaluation Principles:
- ****Standards-Based**:** Adhere strictly to the grading rubric.
- ****Evidence-Based**:** Critique based solely on the text provided; cite specific examples
- ****Encouragement First**:** Adopt an appreciation-based approach (highlight strengths before addressing weaknesses).
- ****Growth-Oriented**:** Focus on actionable steps for improvement.
- ****Task Description**:** Conduct a professional assessment of the submitted Chinese Narrative Essay and generate a comprehensive feedback report.

Task Objectives, Ensure the student achieves the following through your feedback:
- ****Level Awareness**:** Clearly understands their actual proficiency level.
- ****Diagnostic Insight**:** Recognizes specific strengths and critical issues.
- ****Actionable Improvement**:** Receives concrete, feasible suggestions for revision.
- ****Motivation**:** Gains confidence and specific direction for future writing.

(a) The detailed content of System prompt.



<Total Score Evaluation>:

- 8 Points: The essay demonstrates outstanding performance across all three dimensions: prompt adherence, content, and expression. It exhibits a high degree of completion, clear logic, and substantial content, reflecting the author's solid writing proficiency and comprehensive literacy.
- 7 Points: The overall quality is excellent, with outstanding performance in at least two dimensions. The essay is coherent and complete, with only minor areas for improvement.
- 6 Points: The essay is of good quality and generally complete. Performance is stable across most dimensions, though minor issues exist. Overall readability is good.
- 5 Points: The overall quality is average. The essay fulfills the basic task requirements, reaching the standard in at least two dimensions, yet distinct deficiencies remain. The performance is passable but requires improvement.
- 4 Points: The degree of completion is fair. Significant issues exist in at least one dimension. While the essay is finished and structurally intact, improvement is needed in detail and overall cohesion.
- 3 Points: The overall quality is weak, with deficiencies in multiple dimensions (e.g., insubstantial content, weak logic, or inaccurate prompt interpretation). These issues negatively affect the reading experience.
- 2 Points: The essay is problematic, showing insufficiencies in prompt adherence, content, and expression. Overall readability is poor, hindering the effective conveyance of the author's intent.
- 1 Point: The overall quality is poor, characterized by unclear prompt interpretation, incomplete content, and chaotic expression. Comprehension is frequently impeded, indicating a need for fundamental improvement.
- 0 Points: The essay severely deviates from task requirements. The content is irrelevant to the prompt, the structure is chaotic, and the essay fails to form an effective expression.

(b) The detailed content of the prompt for total score evaluation.

Figure 5: The system prompt and total score prompt design for chinese narrative essay evaluation.

B Prompt Template of NarrativeBench

Figure 5, Figure 6 and Figure 7 present the prompt design used for Chinese narrative essay evaluation. Specifically, Figure 5(a) establishes the model's evaluative role as a senior Chinese language teacher and grading committee head, defining its professional persona, assessment principles, diagnostic capabilities, and educational objectives. It emphasizes standards-based, evidence-driven, and growth-oriented evaluation to ensure rigorous yet constructive feedback. Figure 5(b) specifies a total scoring rubric ranging from 0 to 8 points, where each score level corresponds to distinct performance standards across relevance, content, and expression. Moreover, Figure 6(a) illustrates the prompt design for relevance evaluation, while Figure 6(b) presents the prompt design for content evaluation. Figure 7(a) shows the prompt design for expression evaluation, and Figure 7(b) depicts the prompt design for feedback generation.



<Relevance Evaluation>:

- **8 Points**: The essay demonstrates complete command of the prompt. The interpretation is thorough, and the writing remains tightly focused on the theme throughout. Excellent execution.
- **7 Points**: The essay maintains proper control over the prompt. It accurately interprets the task and adheres to the theme, showcasing outstanding prompt analysis skills.
- **6 Points**: The essay demonstrates good command of the prompt with adequate interpretation; however, certain sections do not fully align with the theme.
- **5 Points**: The essay shows a relatively high degree of control and correct interpretation, but fails to maintain consistent thematic focus throughout the writing.
- **4 Points**: The essay demonstrates a basic grasp of the prompt and generally correct interpretation, but the writing contains frequent deviations from the theme.
- **3 Points**: The essay captures the basic requirements of the prompt, yet these requirements are not fully realized or reflected in the actual writing.
- **2 Points**: The essay fails to correctly interpret the prompt; the writing exhibits significant deviations from the intended topic.
- **1 Point**: The essay contains obvious errors in prompt analysis. It fails to adhere to the topic, lacks focus, and demonstrates a lack of basic skills in prompt adherence.
- **0 Points**: The essay is completely irrelevant to the prompt; the interpretation of the topic is entirely incorrect.

(a) The detailed content of the prompt for relevance evaluation.



<Content Evaluation>:

- **8 Points**: The essay features substantial content and rich vocabulary, demonstrating the author's outstanding lexical range and proficiency in diction and syntax.
- **7 Points**: The content is substantial with precise and clear diction, revealing the author's ingenuity and distinct craftsmanship in writing.
- **6 Points**: The essay contains rich content and employs some effective vocabulary and sentences; however, there remains overall room for improvement in lexical variety.
- **5 Points**: The content is adequate and demonstrates good content development and rhetorical effectiveness capable of engaging the reader, though significant room for improvement remains regarding depth and breadth.
- **4 Points**: The essay partially conveys the author's intent, but the substance development is weak. There are noticeable issues with word choice and usage.
- **3 Points**: The essay demonstratesa a basic level of content, yet lacks sufficient richness in content. The vocabulary and sentence structures require further refinement.
- **2 Points**: The expression is relatively dry and fails to demonstrate writing proficiency. The essay lacks necessary vocabulary accumulation and writing proficiency.
- **1 Point**: Basic content ability is deficient, and the content lacks richness. Fundamental improvements are needed in substance.
- **0 Points**: The content is overly simplistic, severely lacking in details and descriptive elements.

(b) The detailed content of the prompt for content evaluation.

Figure 6: The prompts used for relevance and content dimension scoring in Chinese narrative essay evaluation.

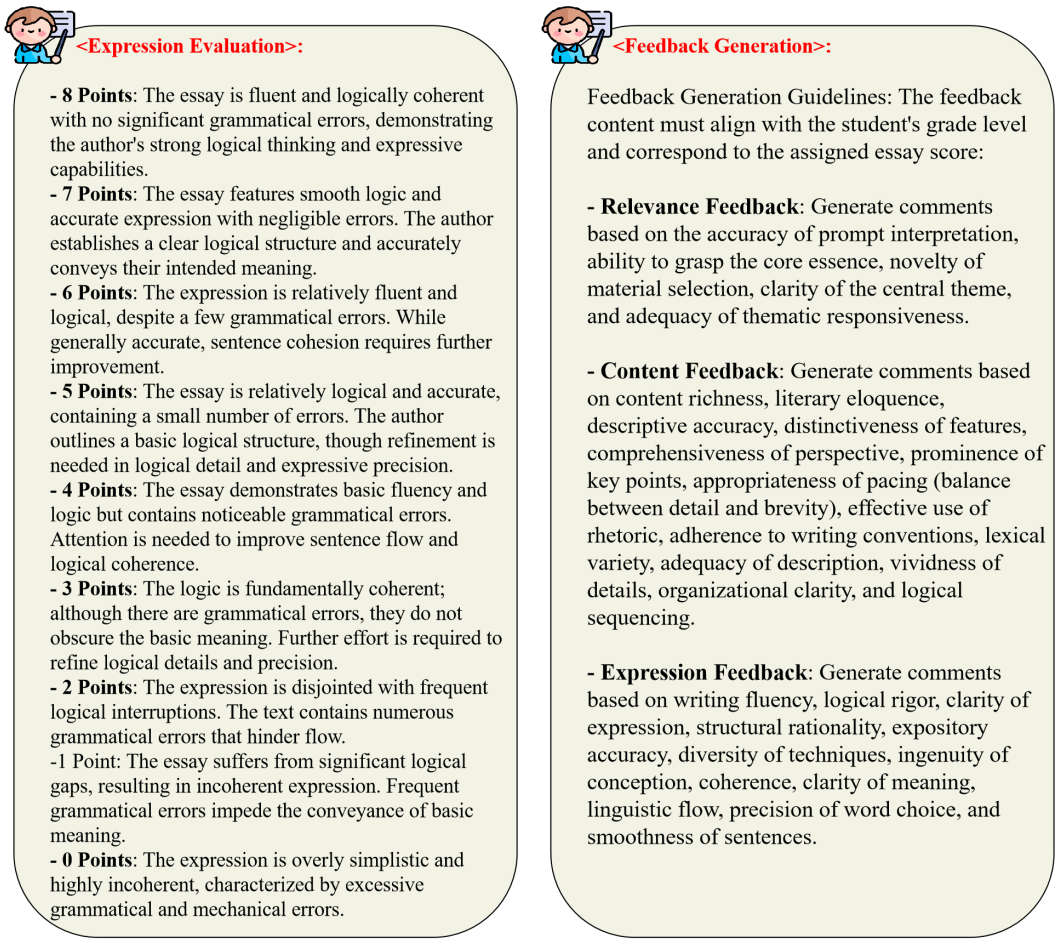
Table 8: The proportion of score differences ≤ 1 between two expert raters for each trait.

Trait	# Essays	Proportion (%)
Total Score	4,216	93.69
Relevance	4,409	97.98
Content	4,382	97.38
Expression	4,398	97.73

C Annotation Details

To evaluate the reliability of the human grading, we analyzed the inter-rater consistency across four key traits: Total Score, Relevance, Content, and Expression. A total of six experts in Chinese Language and Literature were invited, with each essay independently evaluated by two of these six experts. As shown in Table 8, the scoring results demonstrate remarkably high agreement between the paired experts. Specifically, the proportion of score differences falling within the ≤ 1 range exceeds 93% for all dimensions. Relevance achieved the highest consistency at 97.98%, suggesting that experts share a highly unified understanding of whether an essay aligns with the given topic. Even for the most subjective dimension, Total Score, the agreement rate remains robust at 93.69%.

For the remaining cases where score differences exceeded 1 (i.e., $\Delta > 1$), a senior teacher with extensive instructional experience reviewed the essays to adjudicate and determine the final consensus labels. This two-stage process that combines high inter-rater agreement and adjudication of significant



(a) The detailed content of the prompt for expression evaluation.

(a) The detailed content of the prompt for feedback generation.

Figure 7: The prompts used for expression scoring and feedback generation in Chinese narrative essay evaluation.

discrepancies highlights the clarity of our annotation guidelines and confirms the high quality of the final scores used for our experiments.

D Model List

We evaluate a diverse suite of Large Language Models (LLMs), encompassing both open-source and proprietary architectures to ensure a comprehensive assessment of current state-of-the-art capabilities. A detailed inventory of the models, including their parameter scales and accessibility, is provided in Table 9. Our selection spans across globally leading frontier models and representative LLMs, covering a diverse range of parameter scales.

E Closed-Source LLM API Token Consumption and Cost Transparency

For transparency and reproducibility, we report the total API token consumption (including both input and output tokens) for all closed-source LLMs evaluated in this study, while open-source models were deployed locally and therefore incurred no external usage cost. Across models, the total token consumption generally ranged from approximately 11.5 million to 14.5 million tokens, with an average usage of about 13.3 million tokens per model. Specifically, Doubao consumed

Table 9: The information list of LLMs used in this study.

Type	Model	Abbreviations	Type	Parameters	Access	URL
English LLMs	ChatGPT-5.4	GPT-5.4	Closed-source	—	API	OpenAI Link
	Grok-4-reasoning	Grok-4	Closed-source	—	API	Grok Link
	Gemini-3.1-pro-preview	Gemini-3.1-pro	Closed-source	—	API	Google Link
	Llama-4-Scout	Llama-4-Scout	Open-source	109B	Weights	Meta Link
	Claude-Sonnet-4.6	Claude-Sonnet	Closed-source	—	API	Claude Link
Chinese LLMs	Doubao-seed-2.0-pro	Doubao-seed-2.0	Closed-source	—	API	Seed Link
	DeepSeek-V4-pro	DeepSeek-V4	Open-source	862B	Weights	HuggingFace Link
	Qwen-3.6-plus	Qwen-3.6-plus	Closed-source	—	API	Qwen Link
	Qwen-3.5-27B	Qwen-3.5-27B	Open-source	27B	Weights	HuggingFace Link
	Qwen-3.5-9B	Qwen-3.5-9B	Open-source	9B	Weights	HuggingFace Link

14,056,442 tokens, Qwen3.6-plus consumed 13,418,887 tokens, Grok-4-fast-reasoning consumed 11,546,231 tokens, Gemini-3.1-pro consumed 12,652,250 tokens, GPT-5.4 consumed 13,731,508 tokens, and Claude-Sonnet-4.6 consumed 14,492,046 tokens. These statistics provide a practical reference for estimating the computational and financial costs associated with large-scale evaluation using closed-source LLMs. In addition, for the variance and robustness of LLM judgments, please refer to the Appendix H.2.

F Additional Experimental Details

Table 10: Detailed performance of various LLMs on the feedback generation task. The table reports the Relative Preference Strength (S_{total}) across three pedagogical dimensions, alongside semantic similarity metrics.

Model	Preference Strength (S_{total})			Semantic Metrics	
	Relevance	Content	Expression	Cosine Sim.	BERT Score
English LLMs					
GPT-5.4	-1037	2043	1389	0.733 ± 0.068	0.502 ± 0.070
Gemini-3.1-pro	-906	2205	1348	0.712 ± 0.106	0.537 ± 0.093
Claude-4.6-Sonnet	-126	2386	1464	0.805 ± 0.053	0.607 ± 0.058
Llama-4-Scout	-3325	-697	-2184	0.581 ± 0.072	0.493 ± 0.070
Grok-4	-1285	1165	955	0.688 ± 0.115	0.533 ± 0.108
Chinese LLMs					
DeepSeek-V4-pro	221	2383	1386	0.776 ± 0.092	0.569 ± 0.090
Qwen3.6-Plus	-316	2166	1352	0.753 ± 0.055	0.538 ± 0.053
Qwen3.5-27B	-2308	-389	-216	0.598 ± 0.063	0.521 ± 0.088
Qwen3.5-9B	-2470	-523	-762	0.597 ± 0.113	0.515 ± 0.108
Doubao-seed-2.0-pro	109	1983	1205	0.712 ± 0.059	0.539 ± 0.055

G Case Study

To investigate why LLMs consistently underperform on Relevance, we conduct a qualitative error analysis through three representative cases (Tables 11 to 13), each targeting a distinct failure mode: (1) insensitivity to discourse-level thematic alignment due to **linguistic nuance**; (2) conflation of narrative fluency with prompt fulfillment due to **prompt intent miscomprehension**; and (3) severe under-scoring of age-appropriate writing due to **distributional mismatch** between elementary-level text and formal pretraining corpora. These cases demonstrate that the bottleneck of Relevance, Content and Expression stem from multiple separable failure modes, each requiring targeted solutions beyond prompt engineering.

In Case 1 (Table 11), the student essay is substantively on-topic throughout, with an isolated off-topic closing sentence likely resulting from a copy-paste error. The LLM assigns 1/8 despite the expert rating of 3/8, indicating that it disproportionately weights a single anomalous sentence rather than evaluating thematic alignment holistically. Human raters recognize such lapses as minor and score at the discourse level accordingly.

In Case 2 (Table 12), the essay narrates a crab-fishing experience fluently but entirely omits the reflective engagement required by the prompt. The LLM awards 7/8 against an expert score of 3/8, revealing that it rewards narrative coherence independently of whether the deeper communicative intent of the prompt has been fulfilled. This limitation is especially pronounced for prompts where thematic fulfillment requires inferring implicit expectations beyond the literal wording.

In Case 3 (Table 13), the essay fully addresses all prompt dimensions and is rated 8/8 by experts, yet receives only 2/8 from the LLM. The writing employs simple vocabulary and short declarative sentences that are pedagogically appropriate for primary school but appear to trigger inappropriately low scoring. This reflects a fundamental mismatch between LLM evaluative priors calibrated on adult, formal text and the developmental writing standards applied by human raters in educational assessment.

Table 11: Case 1: Over-penalization of a local textual anomaly.

Prompt	漫画的启示*(Inspirations from Comics)*
Essay excerpt	The essay discusses a cartoon about selfishness and extends the theme with a real-life subway example (*"*真是自私自利的'小人"'. . .周围的人看不下去了，就找来秩序员解决此事), maintaining consistent thematic focus. However, the final sentence abruptly deviates: ***"***我相信，只要爷爷想戒烟，就一定可以成功***(I believe that as long as Grandpa wants to quit smoking, he will succeed.)"***
Expert score	3 / 8 — minor deduction for the inconsistent closing sentence; overall thematic alignment preserved.
LLM score	1 / 8
Error type	Over-penalization
Analysis	The LLM disproportionately weights the anomalous final sentence and collapses the overall Relevance score, despite the essay being substantively on-topic. Human raters treat this as an isolated lapse and evaluate holistic relevance accordingly, demonstrating that LLMs lack the global reading strategy necessary for pedagogically consistent scoring.

Table 12: Case 2: Under-penalization of thematic drift.

Prompt	生活万花筒*(Kaleidoscope of Life)*
Essay excerpt	The essay narrates a crab-fishing experience in fluent and coherent prose (*"*根据爷爷教我的方法. . .最终，我钓了许多螃蟹，实现了螃蟹自由的目标*(Following Grandpa's instructions . . . I finally caught many crabs.)"*) but never reflects on the broader significance of the experience, failing to fulfill the reflective intent of the prompt.
Expert score	3 / 8 — penalized for absence of reflection and insufficient engagement with the prompt's thematic requirement.
LLM score	7 / 8
Error type	Under-penalization
Analysis	The LLM conflates surface-level narrative fluency with genuine thematic fulfillment, rewarding coherent event description while overlooking the absence of reflective depth required by the prompt. This reveals a critical limitation: LLMs are prone to rewarding stylistic quality independently of prompt adherence.

Table 13: Case 3: Distributional mismatch with elementary-level writing.

Prompt	我的“自画像” *(My “Self-Portrait”)*
Essay excerpt	The essay fully satisfies all prompt dimensions with well-organized paragraphs covering appearance, personality, hobby, and weakness (“我的性格开朗，积极向上. . . 我还是一个折纸达人. . .我也有不足的地方，就是有点小粗心*(I am outgoing . . . I am also good at origami . . . my weakness is carelessness.)*”) using age-appropriate vocabulary and short declarative sentences.
Expert score	8 / 8 — exemplary self-portrait covering all required dimensions with appropriate grade-level expression.
LLM score	2 / 8
Error type	Distributional mismatch
Analysis	Despite the essay being exemplary for its grade level, the LLM severely under-scores it. Simple vocabulary and short sentence structures — pedagogically appropriate for primary school writing — are likely penalized due to a distributional mismatch between elementary-level text and the formal, high-register corpora that dominate LLM pretraining[Touvron et al., 2023], revealing a fundamental misalignment between LLM evaluative priors and developmental writing standards.

Table 14: The analysis of bias scores for the two judge LLMs (DeepSeek-V4-pro and Gemini-3.1-pro). The reported values are net preference scores (see Eq. (2) on page 4).

Model	Judge LLM(s)	Relevance	Content	Expression
DeepSeek-V4-pro	AVG (DeepSeek-V4-pro + Gemini-3.1-pro)	221	2383	1386
DeepSeek-V4-pro	DeepSeek-V4-pro	229	2390	1372
DeepSeek-V4-pro	Gemini-3.1-pro	213	2376	1400
DeepSeek-V4-pro	Kimi-K3	220	2385	1383
Gemini-3.1-pro	AVG (DeepSeek-V4-pro + Gemini-3.1-pro)	-906	2205	1348
Gemini-3.1-pro	Gemini-3.1-pro	-899	2181	1346
Gemini-3.1-pro	DeepSeek-V4-pro	-913	2229	1350
Gemini-3.1-pro	Kimi-K3	-909	2213	1345

H Bias Analysis of LLMs

H.1 Self-Preference Bias Analysis of LLMs

Using an evaluated model as a judge may introduce self-preference bias. Although our original averaging over DeepSeek-V4-pro and Gemini-3.1-pro reduces the influence of either single judge, it does not by itself rule out this bias. We therefore conducted an additional analysis using Kimi-K3, which is not included in the evaluated model set, under the same evaluation protocol. The results are shown in Figure 14.

We added Kimi-K3, which is not included in the evaluated model set, as an independent third judge. Averaging all three judges (DeepSeek-V4-pro, Gemini-3.1-pro and Kimi-K3) yields Relevance, Content, and Expression scores of (220.7, 2383.7, 1385.0) for DeepSeek-V4-pro and (-907.0, 2207.7, 1347.0) for Gemini-3.1-pro, closely matching the original two-judge results (DeepSeek-V4-pro and Gemini-3.1-pro). More importantly, after excluding each model's self-judgment, averaging the two remaining non-self judges yields (216.5, 2380.5, 1391.5) for DeepSeek-V4-pro and (-911.0, 2221.0, 1347.5) for Gemini-3.1-pro, without changing any qualitative conclusion. In particular, DeepSeek-V4-pro retains a positive Relevance score when judged only by Gemini-3.1-pro and Kimi-K3, supporting the observation in Lines 261–262.

H.2 Result Variance and Robustness of LLMs

In our main experiments, all open- and closed-source LLMs were evaluated using the same decoding configuration with a temperature of 0.1. This low-temperature setting reduces stochastic variation and improves comparability across models, and guarantees reproducibility to some extent.

Table 15: Analysis of variance and stability in LLM-as-a-Judge scoring across three runs.

Model	Setting	Run 1 Avg. QWK	Run 2 Avg. QWK	Run 3 Avg. QWK	Mean ± SD
DeepSeek-V4-pro	Zero-shot	0.2880	0.2892	0.2873	0.2882 ± 0.0010
DeepSeek-V4-pro	Few-shot	0.3030	0.3062	0.3008	0.3033 ± 0.0027
Gemini-3.1-pro	Zero-shot	0.3030	0.3071	0.3012	0.3038 ± 0.0030
Gemini-3.1-pro	Few-shot	0.3060	0.3080	0.3049	0.3063 ± 0.0016
GPT-5.4	Zero-shot	0.2850	0.2874	0.2836	0.2853 ± 0.0019
GPT-5.4	Few-shot	0.2910	0.2954	0.2882	0.2915 ± 0.0036

Table 16: Agreement between expert preferences and LLM judgments (E.A./QWK). Each table entry reports Exact Agreement (E.A.) followed by QWK. Human–Human denotes agreement between the two teachers before adjudication, whereas LLM–Human denotes agreement between each LLM judge and the adjudicated human consensus. The Overall results are computed by pooling all 300 dimension-level comparisons.

Dimension (E.A./QWK)	N	Human–Human	DeepSeek–Human	Gemini–Human
Relevance	100	0.910 / 0.914	0.850 / 0.864	0.840 / 0.859
Content	100	0.940 / 0.946	0.880 / 0.891	0.870 / 0.889
Expression	100	0.950 / 0.952	0.890 / 0.895	0.880 / 0.888
Overall	**300**	**0.933 / 0.937**	**0.873 / 0.883**	**0.863 / 0.878**

To directly address concerns regarding result variance and robustness (each LLM tested with only one inference pass per essay,), we therefore conducted a repeated-inference robustness analysis on a stratified subset of 300 essays using three representative models: DeepSeek-V4-pro, Gemini-3.1-pro, and GPT-5.4. Each model was independently run three times under identical zero-shot and few-shot settings.

Across the repeated runs, the standard deviation of Avg. QWK ranges from 0.0010 to 0.0036 (see Table 15), indicating limited run-to-run variation. Moreover, the relative ordering of the three models remains unchanged across all runs, and few-shot prompting consistently outperforms the corresponding zero-shot setting. These results indicate that inference stochasticity under our low-temperature configuration has only a limited effect on the aggregate QWK results and does not alter the principal conclusions.

I LLM–Human Preference Alignment Analysis

To ensure consistency between expert preferences and LLM judgments, thereby determining whether the pairwise judging protocol is reliable (lines 131–144), We conducted a human spot-check on 300 dimension-specific pairwise comparisons, with 100 comparisons sampled from each of Relevance, Content, and Expression. The samples were stratified across evaluated models and essay prompts.

Two Chinese-language teachers who had not participated in the original feedback annotation independently assessed each comparison using the same three-way preference labels (-1, 0, and 1) and evaluation criteria as the LLM judges, including accuracy, relevance, and pedagogical usefulness. Model identities were anonymized, and the A/B presentation order was randomized. Disagreements between the two teachers were adjudicated by a third senior teacher to obtain the final human-consensus labels.

As shown in Figure 16, the two initial teachers achieved an overall Exact Agreement of 93.3% and a QWK of 0.937, indicating high inter-rater reliability. Compared with the adjudicated human consensus, DeepSeek-V4-pro achieved an Exact Agreement of 87.3% and a QWK of 0.883, while Gemini-3.1-pro achieved an Exact Agreement of 86.3% and a QWK of 0.878. Importantly, these metrics measure the agreement of each LLM judge with the human consensus independently and do not rely on majority voting or any other ensemble decision.

The dimension-specific results further show that the overall agreement is not driven by a single dimension. Both LLM judges remain highly aligned with expert preferences across Relevance, Content, and Expression, with QWK values ranging from 0.859 to 0.895. Agreement is slightly lower

for Relevance than for Content and Expression, consistent with the greater evaluation difficulty of this dimension, but it still demonstrates strong alignment with human judgments. Overall, this human spot-check provides direct support for the reliability of our LLM-based pairwise judging protocol. We will add the human sampling and annotation procedure, agreement analysis, and corresponding limitations to the revised manuscript.

NeurIPS Paper Checklist

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   Question: Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope?

   Answer: [Yes]

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2. **Limitations**

   Question: Does the paper discuss the limitations of the work performed by the authors?

   Answer: [Yes]

   Justification: We discuss the limitations of this work in the Limitation section. Due to the high evaluation cost, and consistent with existing LLM evaluation studies, we conducted only a single inference run for each LLM via API. To minimize the impact of sampling randomness and improve the stability of the evaluation results, we uniformly set the temperature parameter to 0.1 for all models during inference.

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Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?

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Justification: This paper does not include theoretical results, formal theorems, or mathematical proofs. Instead, it focuses on benchmark evaluation of LLMs for Chinese narrative essay assessment.

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Question: Does the paper fully disclose all the information needed to reproduce the main experimental results of the paper to the extent that it affects the main claims and/or conclusions of the paper (regardless of whether the code and data are provided or not)?

Answer: [Yes]

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At the same time, we acknowledge several potential negative societal impacts. Current LLMs still exhibit substantial gaps compared to expert human evaluators, especially in relevance assessment and fine-grained semantic understanding. Incorrect scoring or misleading feedback could unfairly affect students if such systems are directly deployed in high-stakes educational settings. In addition, biases inherited from training data or model preferences may lead to inconsistent evaluations across writing styles or student groups. Therefore, we emphasize that current LLMs should be viewed as assistive tools rather than replacements for human teachers. Our benchmark is intended to support more transparent, reliable, and carefully validated educational AI systems.

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